

ASSESSMENT CO1 AT3

OUTPUT PREDICTION

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Reg no:192525194

Course Code: CSA0312

Course Name: Data Structures

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=365`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the course "DS PROGRAMMING". The user's name "Bedudoori Siva Parvathi" and registration number "192525194RC" are visible in the top right corner.

On the left sidebar, there is a list of questions: "Reverse a string" (1 mark), "Q3 Searching" (1 mark), "Q4 Searching" (1 mark), "Q5 Array" (1 mark), "Q6 Merging" (1 mark), "Q7 Largest and Smallest element" (1 mark), and "Q8 Second Largest Element".

The main area contains a C program for reversing an array:

```
1 #include <stdio.h>
2 int main() {
3     int n, i, temp;
4     printf("Enter the number of elements: ");
5     scanf("%d", &n);
6     int arr[n];
7     printf("Enter %d elements:\n", n);
8     for (i = 0; i < n; i++) {
9         scanf("%d", &arr[i]);
10    }
11    for (i = 0; i < n / 2; i++) {
12        temp = arr[i];
13        arr[i] = arr[n - 1 - i];
14        arr[n - 1 - i] = temp;
15    }
16    printf("Reversed array:\n");
17    for (i = 0; i < n; i++) {
18        printf("%d ", arr[i]);
19    }
20    printf("\n");
21    return 0;
22 }
```

On the right side, there is an input field with the value "5" and a list of numbers "1 2 3 4 5". Below this, the "Output" section shows the program's execution:

```
Enter the number of elements:
Enter 5 elements:
Reversed array:
5 4 3 2 1
```

The Windows taskbar at the bottom shows the system clock as 02:10 PM on 25-07-2026.



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Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7
Largest and Smallest element
1 mark

Q8
Second Largest Element
1 mark

Q9
Occurrence of an element
1 mark

Your Code

```
1 #include <stdio.h>
2 int main()
3 {
4     char s[50];
5     int i, n = 0, flag = 1;
6     printf("Enter string: ");
7     scanf("%s", s);
8     while (s[n] != '\0')
9         n++;
10    printf("Reverse: ");
11    for (i = n - 1; i >= 0; i--)
12        printf("%c", s[i]);
13    for (i = 0; i < n / 2; i++)
14    {
15        if (s[i] != s[n - 1 - i])
16        {
17            flag = 0;
18            break;
19        }
20    }
21    if (flag == 1)
22        printf("\nPalindrome");
23    else
24        printf("\nNot Palindrome");
25    return 0;
26 }
```

Input

hello

Output

Security Error: Disallowed code pattern detected.



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Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7
Largest and Smallest element
1 mark

Your Code

```
1 #include <stdio.h>
2 int main(){
3     int n,a[100],x,i;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)
6         scanf("%d",&a[i]);
7     scanf("%d",&x);
8     for(i=0;i<n;i++){
9         if(a[i]==x){
10             printf("found at position %d",i+1);
11             return 0;
12         }
13     }
14     printf("Not found");
15 return 0;
16 }
```

Run Save Submit

Input

5
10 20 30 40 50
30

Output

found at position 3

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Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7

Write a C program for Count the Occurrences of an Element

Your Code

```
1 #include <stdio.h>
2 int main()
3 {
4     int a[20], n, x, i, c=0;
5     scanf("%d", &n);
6     for(i=0; i<n; i++)
7         scanf("%d", &a[i]);
8     scanf("%d", &x);
9     for(i=0; i<n; i++)
10         if(a[i]==x)
11             c++;
12     printf("occurences=%d", c);
13     return 0;
14 }
```

Run Save Submit

Input

5
10 20 10 30 10
10

Output

occurences=3

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Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7
Largest and Smallest element
1 mark

Q8
Second Largest Element
1 mark

Q9
Occurrence of an elemnet
1 mark

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20], n, x, l=0, h, mid, i;
5     scanf("%d", &n);
6     for (i=0; i<n; i++)
7         scanf("%d", &a[i]);
8     scanf ("%d", &x);
9     h=n-1;
10    while (l<=h)
11    {
12        mid=(l+h)/2;
13        if (a[mid]==x)
14        {
15            printf("found at %d", mid+1);
16            return 0;
17        }
18        if (x>a[mid])
19            l=mid+1;
20        else
21            h=mid-1;
22    }
23    printf("not found");
24 }
```

Run Save Submit

Input

5
10 20 30 40 50
40

Output

found at 4

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▼

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merging

1 mark

✓

Q7

Largest and Smallest element

1 mark

✓

Your Code

```
1 #include <stdio.h>
2 int main () {
3     int a[100], n, i, max, min;
4     scanf("%d", &n);
5     for (i=0; i<n; i++)
6         scanf("%d", &a[i]);
7     max=min=a[0];
8     for (i=1; i<n; i++) {
9         if (a[i]>max)
10             max=a[i];
11         if (a[i]<min)
12             min=a[i];
13     }
14     printf("largest =%d \n", max);
15     printf("smallest =%d", min);
16     return 0;
17 }
```

▶ Run

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➡ Submit

Input

5
10 20 5 30 15

Output

largest =30
smallest =5

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▼

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merging

1 mark

✓

Q7

Largest and Smallest element

1 mark

✓

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[10], b[10], c[20], n1, n2, i;
4     scanf("%d", &n1);
5     for (i=0; i<n1; i++)
6         scanf("%d", &a[i]);
7     scanf("%d", &n2);
8     for (i=0; i<n2; i++)
9         scanf("%d", &b[i]);
10    for (i=0; i<n1; i++)
11        c[i]=a[i];
12    for (i=0; i<n2; i++)
13        c[n1+i]=b[i];
14    for (i=0; i<n1+n2; i++)
15        printf("%d ", c[i]);
16    return 0;
17 }
```

▶ Run

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Input

3
10 20 30
4
40 50 60 70

Output

10 20 30 40 50 60 70



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Q2 Reverse a string
1 mark

Q3 Searching
1 mark

Q4 Searching
1 mark

Q5 Array
1 mark

Q6 Merging
1 mark

Q7 Largest and Smallest element
1 mark

Q8 Second Largest Element
1 mark

Q9 Occurrence of an element
1 mark

Your Code

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[20],n,x,i,first=-1,last=-1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d",&x);
9     for(i=0;i<n;i++)
10    {
11        if(a[i]==x)
12        {
13            if(first==-1)
14                first=i+1;
15            last=i+1;
16        }
17    }
18    if(first==-1)
19        printf("Not Found");
20    else
21        printf("First=%d\nLast=%d",first,last);
22    return 0;
23 }

```





Input

```

7
10 20 30 20 40 20 50
20

```

Output

```

First=2
Last=6

```

ff



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Q1 Reverse a list
1 mark

Q2 Reverse a string
1 mark

Q3 Searching
1 mark

Q4 Searching
1 mark

Q5 Array
1 mark

Q6 Merging
1 mark

Q7 Largest and Smallest element
1 mark

Q8 Second Largest Element
1 mark

Write a C program for Find the Second Largest Element

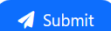
Your Code

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[20],n,i,first,second;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     first=second=a[0];
9     for(i=1;i<n;i++)
10    {
11        if(a[i]>first)
12        {
13            second=first;
14            first=a[i];
15        }
16        else if(a[i]>second && a[i]!=first)
17            second=a[i];
18    }
19    printf("Second Largest = %d",second);
20    return 0;
21 }

```





Input

```

5
10 20 30 80 100

```

Output

```

Second Largest = 80

```

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▼

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merging

1 mark

✓

Q7

Largest and Smallest element

1 mark

✓

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],n,i;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n-1;i++)
9     {
10         if(a[i+1]-a[i]!=1)
11         {
12             printf("Missing Element = %d",a[i]+1);
13             return 0;
14         }
15     }
16     printf("No Missing Element");
17     return 0;
18 }
```

▶ Run

💾 Save

🚀 Submit

Input

5
1 2 3 5 6

Output

Missing Element = 4